KOMA

Izpiti: 25.1 (21.2)
4.2

Kolokviji:
1) torek 24.11 ob 17.30
2) torek 12.1 ob 17.30

1) Naj bo k polje in $k[V]$ $k[X]$-algebra podana s $k[X] \to k[V]$

$X \mapsto V^2$

$\sum \mu_i X^i \mapsto \sum \mu_i V^{2i}$

$k[V]$ je kolobar in $k[X]$-modul

$\lambda(x \cdot y) = (\lambda x) y = x(\lambda y)$

a) pokaži da je $k[V]$ prosti modul $k[X]$ in $k[V] \underset{\text{vig}}{\cong} \frac{k[X][T]}{(T^2 - X)}$

b) Pokaži $B := k[V] \otimes_{k[V]} k[V] \overset{\text{kol}}{\cong} \frac{k[V,W]}{(V^2 - W^2)}$

c) Če $\operatorname{char} k \neq 2$ pokaži: da B ni cel kolobar, ima nima nilpotenta in $B \cong \{(f,g) \in k[V] \times k[V] ; f(0) = g(0)\}$

d) Naj bo $\operatorname{char} k = 2$. Poišči nilpotent (neničelni) v B

a) $\lambda_0 + \lambda_1 v + \dots + \lambda_n v^n$

$x \cdot 1 = v^2$

$x \cdot v = v^3$

$x \cdot v^n = v^{n+2}$

$(1+x) \cdot 1 = 1 + v^2$

$\{1, v\}$

$\lambda_0 + \lambda_1 v + \dots \lambda_n v^n =$

$(\lambda_0 + \lambda_2 v^2 + \dots) + v(\lambda_1 + \lambda_3 v^2 + \dots) =$

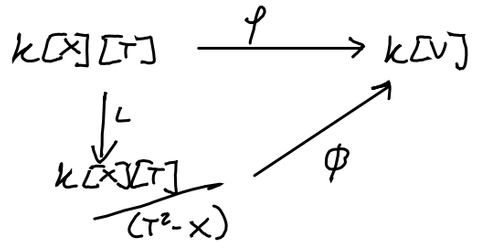
$= (\lambda_0 + \lambda_2 x + \dots) \cdot 1 + (\lambda_1 + \lambda_3 x + \dots) v$

$$k[v] \simeq \frac{k[x][T]}{(T^2 - x)}$$

$a_0 + a_1 v \longmapsto a_0 + a_1 T$

Ali to ohranja množenje? ...

2. Način



$\varphi: \quad T \longmapsto v$

$\quad\quad x \longmapsto v^2$

ohranja strukturo modula

$\sum_i \left(\left(\sum_j \mu x^j \right) v_i T^j \right) \longmapsto \left(\sum \mu v^{2j} \right) \left(\sum v_i v^j \right)$

$= \sum_{i,j} \mu_i x^j v_i T^j \longmapsto \sum \mu_i v_i v^{2j} w^j$ " ✓

Kaj je jedro?

$\ker \varphi = ?$

$(T^2 - x^k) \longmapsto v^2 - v^2 = 0$ to je v jedru

$\sum v_{jk} x^j T^k \longmapsto \sum v_{jk} x^j v^k =$

$= p(x) \cdot 1 + q(x) v$

$= \underbrace{\sum \lambda_i v^{2i}}_{=0} + \underbrace{\sum \gamma_j v^{2j+1}}_{=0}$

b)

$$k[V]\otimes \frac{k[X][T]}{(T-X)} \cong \frac{(k[V]\otimes k[X])(T)}{(T^2-X)} \cong$$

$$\cong \frac{k[V][T]}{(T^2-X)} \cong k[V]\frac{[T]}{(T^2-V)}$$

2. način

$$\varphi: k[V]\otimes k[V] \longrightarrow \frac{k[V,W]}{(V^2-W^2)}$$

$$\sum \mu_i (\rho_i(V)\otimes \xi_i(W)) \longmapsto \sum \mu_i \rho_i(V)\xi_i(W)$$

Univerzalna lastnost za algebre

$$\begin{array}{ccc} A & \xrightarrow{\hom_\varphi} & C \\ B & \xrightarrow{\hom_\psi} & \end{array} \quad \rightsquigarrow \quad A\otimes B \longrightarrow C$$

$$a\otimes b \longmapsto \varphi(a)\psi(b)$$

+ pogoj komutativnosti

Preslikavi sta identiteti

$$k[V] \longmapsto \frac{k[V,W]}{(V^2-W^2)}$$

$$V \longmapsto V$$

$$V \longmapsto W$$

To je preslikava v eno smer

Druga smer:

$$\Psi\ k[V,W] \longrightarrow k[V]\otimes k[V]$$

$$\sum_{ij} \lambda_{ij} V^i W^j \longmapsto \lambda_{ij}(V^i\otimes W^j)$$

$$V \longmapsto (V\otimes 1)$$

$$W \longmapsto (1\otimes W)$$

Po univerzalni lastnosti je to homomorfizem

$(V^2-W^2) \in \ker \Psi$

$$(V^2\otimes 1) - (1\otimes W^2) = ((X-1)-(X-1))(1\otimes 1) = 0$$

$\Rightarrow$ to je homomorfizem

$$\frac{k[V,W]}{(V^2-W^2)} \longrightarrow k[V]\otimes k[V]$$

Ali je to izomorfizem?

$\Phi\circ\overline{\Psi} = id$

Vstavimo v in w (če je na generatorjih id je povsod id

$$(\Phi\circ\overline{\Psi})(V) = \Phi(V\otimes 1) = V$$

$$\overline{\Psi}(\Phi(f\otimes g)) = \Psi(f(V)\cdot g(W)) = (f\otimes g)$$

c)

$$\mathcal{B} = k[V] \otimes k[V] = \frac{k[v,w]}{(v^2-w^2)}$$

$$\mathcal{B} \cong \{(f,g);\ h[V] \times h[V];\ f(0) = g(0)\}$$



B_cel

- komutativnost ✓
- obrnljivost ?

  nima deliteljev ničla

$$\frac{k[v,w]}{(v-w)(v+w)} \longrightarrow$$

crt: I, J ideala, ki sta si tuja v algebri A

$I + J = A$

Potem $A/_{IJ} \cong A/_I \times A/_J$

Al: sta tuja

$$I + J = p(x)(v-w) + g(x)(v+w) =$$

$$= (p(x)+g(x))(v) + (p(x)-g(x))w = (v,w)$$

Boj splošno

$$A/_{I \cap J} \cong \{(f,g) \in A/_I \times A/_J;\ f \equiv g \bmod (I+J)\}$$

$$A \longrightarrow C \qquad \text{surjektivnost}$$

$$a \mapsto (a \bmod I, a \bmod J)$$

$I \cap J = (v^2 - w^2)$

$\supseteq$ ✓

$\subseteq$ $f \in I \cap J \Rightarrow (v-w) \mid f$ in $(v+w) \mid f$

ker sta nerazcepna in nessociirana

$\underset{UFD}{\Rightarrow}$ bo $(v-w)(v+w) \mid f$

$$\Rightarrow \cong \{(f,g) \in \frac{k[v,w]}{(v-w)} \times \frac{k[v,w]}{(v+w)}\}$$

$$\frac{k[u,v]}{(v \pm w)} \cong k[y]$$

$f(v,w) \longmapsto f(y, \mp y)$ ta je surjektivna

$f(0) = g(0)$

$\lambda_0 = \gamma_0 \bmod (v,w)$

⋮

$R = \mathcal{P}(X)$ enota

(2.) $A + B = (A - B) \cup (B - A)$ $\emptyset$

$A \cdot B = A \cap B$ $\mathcal{U}$

a) $I \circ R \Leftrightarrow$ zapite za podmnožice in končne unije

$I \subseteq R$

$I \neq \emptyset$

($\Leftarrow$)

$A, B \in I$ –

$A + B = (A - B) \cup (B - A) \Rightarrow$ ✓

($A - B \subseteq A$, $B - A \subseteq B$)

$A \cdot B = A \cap B \subseteq A$

($\Rightarrow$) Naj bo $A \in I$ $B \subseteq A$

$\underline{B = A \cap B = A \cdot B \in I}$

$A, B \in I$

$A \cup B = \underbrace{(A - B) \cap (B - A)}_{\in I = C} \cup \underbrace{(A \cap B)}_{\in I} =$

$C \cup (A \cap B) =$

$(C - (A \cap B)) \cup ((A \cap B) - C) = A \cap B + C \in I$

b) Dokaži $\langle A \rangle$ in $\langle A, B \rangle$

$$\mathcal{R}\cdot A = \{ B \cap A ; B \in \mathcal{R} \} = \mathcal{P}(A)$$

$\langle A, B \rangle$

$A \cup B \in \langle A, B \rangle \Rightarrow \langle A, B \rangle \subseteq \mathcal{P}(A \cup B)$

- $\mathcal{P}(A \cup B) \subseteq \langle A, B \rangle$

$C \in \mathcal{P}(A \cup B) = (\underbrace{C \cap A}_{\in I}) \cup (\underbrace{C \cap B}_{\in I}) =$

$= ((C \cap A) - (C \cap B)) \cup (C \cap B) - (C \cap A)) \cup (C \cap A \cap B)$

$\langle A, B \rangle \subseteq \mathcal{P}(A \cup B)$

$(C \cap A) + (C \cap B) \subseteq A \cup B \rightsquigarrow C \in \mathcal{P}(A \cup B)$

$\langle A_1, \ldots A_n \rangle = \langle A_1 \cup A_2, A_3, A_4 \ldots A_n \rangle = \ldots = \langle A_1 \cup A_2 \cup A_3 \ldots \rangle$

c) X neskončna

$\mathcal{I} = \{ A \subseteq X$ je končna $\}$ je ideal ki ni končno generiran

Recimo da je končno generiran.
Potem $\mathcal{I} = \mathcal{P}(A_1 \cup A_2 \cup \ldots \cup A_n) = \mathcal{P}(A)$

A je končna, X pa ima neskončno enojce

$\Downarrow$ $\mathcal{P}(A)$ je končna)

$\Downarrow$ $\mathcal{I}$ je neskončen

Ali je to ideal po točki a) ja

Boolov kolobar ... vsi so idempotentni
$A \cdot A = A$

Vsi Boolov kolobar $\cong P(S)$ za neko končno S
končen

$a \cdot a = a$ $R = \{a_1 \dots a_n\}$ je boolov

~~Velja: $A \cap A = A \Rightarrow R = P(X)$ je boolov~~

$P(S)$ ima n elementov
želimo $n = 2^k$
$n = 2^k$

$\exists\, \forall a_i \in A.\ A \cong A/(1-a) \times A/a$
$R \cong R \cdot e \times R(1-e) \cong$
obe sta boolova s končno elementi
indukcija po moči $2^{n_1} \cdot 2^{n_2} = 2^{n_1+n_2}$ ✓
$n = 2 \dots n = 2^\Phi$ ✓

$R \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \dots \times \mathbb{Z}_2$

Imamo S s k elementi. $S = [k]$
$P(S)$

$R = \mathbb{Z}_2 \times \dots \times \mathbb{Z}_2 \cong S(k)$ je končen boolov kolobar (je izomorfen $\mathbb{Z}_2 \times \dots \times \mathbb{Z}_2$)

$\mathbb{Z}_2^{\mathbb{N}}$
$R = \{a \in \mathbb{Z}_2^{\mathbb{N}}$ končno mnogo a_i neničelnih$\}$ neskončno mnogo enk
$R = \bigoplus_{\mathbb{N}} \mathbb{Z}_2$

↑ števno neskončen $P(S)$ je ali končen ali nešteven neskončen

$A[[X]] = \{ f = \sum_{n \geq 0} a_n X^n \quad a_n \in A \}$

a) f obrnljiv v $A[[X]] \Leftrightarrow a_0$ obrnljiv v A

Ne bo blo

b) Naj bo K polje

$I \trianglelefteq K[[X]] \quad I \neq 0 \Rightarrow I = \langle X^n \rangle$

Naj bo I neničeln ideal

$$\sum a_n X^n \cdot \sum b_k X^k = \sum_n \sum_{k=0}^{n} a_{n-k} b_k X^n$$

$\langle X^n \rangle = \{ p(x) X^n \}$

x je najmanjši n da $a_n \neq 0$ za

$\sum a_k X^k$ lahko zapišemo kot $X^n \sum_{k \geq n} a_{k-n} X^k$

$\Rightarrow I \subseteq \langle X^n \rangle$

$X^n \in I$

$X^n \sum_{k=0}^{\infty} a_{k+n} X^k \qquad a_{k+n} \neq 0$

ker a_n ni 0 je dobljen

c)

$\mathrm{char}\, k \neq 2 \quad 1+x = f^2 \quad f \in k[[x]]$

$$\left(\sum a_n x^n\right)^2 = \sum_{n=0}^{\infty} \sum_{k=0}^{n} a_{n-k} b_k x^n = 1 + x$$

$$(1+x)^{\frac{1}{2}} = \sum a_0 x^n \quad \text{taylorjeva vrsta}$$

$$= \sum \binom{\frac{1}{2}}{k} x^k$$

$$\binom{\frac{1}{2}}{k} = \frac{\overbrace{\frac{1}{2} \cdot \left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)\ldots\left(\frac{1}{2}-k+1\right)}^{k}}{k!} = \ldots$$

$$= \frac{2^k}{k!}(-1)(-3)(-5)\ldots(1-2k) =$$

$$\frac{(2k-1)!!\,(-2)^k}{k!}$$

Kaj pa če $\mathrm{char}\, k \mid k$

$$a_0 b_0 = 1$$

$$a_0 b_1 + a_1 b_0 = 1$$

oziroma je 0 če $a_0 \neq 0$ je to vredu

$$a_0^2 = 1 \qquad \text{Recimo da } a_0 = 1$$

$$2 a_0 a_1 = 1 \qquad \Rightarrow a_1 = \frac{1}{2}$$

$$a_1^2 + 2 a_2 a_0 = 0 \qquad \Rightarrow a_2 = -\frac{1}{2^3}$$

$$2 a_3 a_0 + 2 a_1 a_2 \Rightarrow a_3 = -\frac{a_1^3 a_2}{a_0} = \frac{1}{2}$$

Kaj če char 2 ?

$$x(a_1 a_0 \cdot 2) = 0 \neq 1$$

③ e, e' idempotenta v R

$(e)^2 = (e)$

$(e) = Re$

$(e)^2 = (Re)(Re) =$

$= \{ae \cdot be\} = \{ab e^2\} = \{ab e\} \in (e)$

$(e) \subseteq (e)^2$

obvs. $e = e \cdot e$

$(a)^2 = (a) \Rightarrow a = (f)$ za nek f

$(e) = (e') \Rightarrow e = e'$

$Ra = Ra \cdot Ra = \{ba \cdot ca\} = \{bc \cdot a^2\}$

$(a)^2 = (a^2)$

$(a) = (a^2)$

$Ra = Ra^2$

$ba = ca^2$

$ka^2 = a \quad / k$

$(ka)^2 = ka \Rightarrow ka$ je idempotent

$(ka)R = (R \cdot k) \cdot a \subseteq R \cdot a$

$a = ka^2 \Rightarrow a \in R \cdot k \cdot a \Rightarrow Ra \subseteq R \cdot ka$

Rezimo da

$(e) = (e')$

$Re = Re'$

$e' = ke$

$e = ce'$

$e = cke$

a) $\mathbb{F}_p[x] = \mathbb{Z}_p[x]$

$\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$

$\varphi \; \mathbb{Z}[x] \longrightarrow \mathbb{F}_p[x]$
$\sum a_i x^i \mapsto \sum (a_i \bmod p)\, x^i$

$\ker \varphi = \{p \cdot f(x);\ f(x) \in \mathbb{Z}\} = p \cdot \mathbb{Z}[x] = (p)$

$\mathbb{F}_p[x] \cong \frac{\mathbb{Z}[x]}{p\mathbb{Z}[x]}$

$\mathbb{Z}[x] \xrightarrow{\psi} \mathbb{Z}[i]$
$x \longmapsto i$
$\sum a_i x^i = \sum a_j i^j$ To je eval

$\ker \varphi = \{f(x);\ f(i) = 0\}$
$= (x^2+1)$

$f(x) = p(x^4) + x g(x^4) + x^2 r(x^4) + x^3 s(x^4)$
$f(1) = p(1) + i\, g(1) - r(1) - i\, s(1)$

$\mathbb{Z}[x]/(x^2+1) \xrightarrow{\psi} \mathbb{Z}[i]$

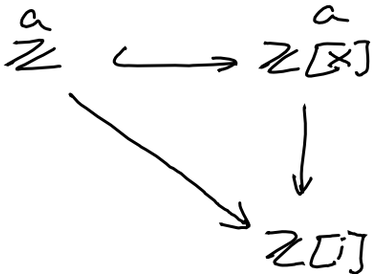
če je $\ker \psi = (x^2+1)$
potem je izo izbrano

$f = \sum a_j x^j + (x^2+1)$

$f = \sum b_0 + b_1 x + (x^2+1)$

$\psi(f) = \sum b_0 + b_1 i$
$= 0 \iff b_0 = 0 = b_1 = 0 \iff f = 0$

$x^3 = x^2 \cdot x =$
$= ((x^2+1) - 1) \cdot x =$
$0 - x = -x$



$$\mathbb{F}_p/(x^2+1) \cong \mathbb{Z}[i]/(p)$$

$$\mathbb{F}_p[x]/(x^2+1) \cong \mathbb{Z}[x]/(p)/(x^2+1) \cong \mathbb{Z}[x]/(p, x^2+1) \cong$$

$$\cong \mathbb{Z}[x]/(x^2+1)/(p) \cong \mathbb{Z}[i]/(p)$$

Kdaj je -1 kvadrat v $\mathbb{Z}_p$?

$$-1 = a^2 \Rightarrow a^2 = p-1 \bmod p$$

-1 ni kvadrat v $\mathbb{F}_p \Leftrightarrow$

$$\mathbb{F}_p^\times = \mathbb{Z}/(p-1) = \langle \zeta \rangle = \{1, \zeta, \dots \zeta^{p-2}\}$$

$$x^2 = -1 \quad \zeta^{k2} = -1 = \zeta^{\frac{p-1}{2}}$$

$$2k = \frac{p-1}{2} \bmod p-1$$

$$k = \frac{p-1}{4} \quad \vee \quad k = -\frac{p-1}{4}$$

$$p-1 = 0 \bmod 4$$

$$\Rightarrow (\zeta^{\frac{p-1}{4}})^2 = -1$$

Za kolobarje določi maksimalne ideale in enote
b) kolik idealov ki niso max imajo

$\mathbb{Z}/(2)[X] = \mathbb{Z}_2[X]$

f - nerazcepen

$X+1, X, X^2+X+1$ so edini nerazcepni

enote: 1 $\quad \mathbb{Z}/(2)[X]^\times = \{1\}$

M max $\Rightarrow$ M prime

$(f) = (gh) \Rightarrow \dfrac{\mathbb{Z}/(2)[X]}{(h)}$ cel? NE

$gh = 0$

$\mathbb{R}[[X]]$

units: tisti kjer je konstantni člen obrnljiv

vsi ideali: (0), $\langle X^n \rangle$ $n \in \mathbb{N}_0$ (projekcije vaje)

maximalni: $\langle X \rangle$

praideali: $\langle 0 \rangle, \langle X \rangle$

$\mathbb{Z}[X]$:

$\mathbb{Z}[X]^\times$ = prosti člen je 1 ali -1

maximalni: $\langle X, p \rangle$ $p \in \mathbb{P}$

I nek max ideal $X \in I$

$I + \langle X \rangle \neq \mathbb{Z}[X]$

Recimo da $I + (X) = \mathbb{Z}[X]$

$1 = f + X \cdot g(X)$

f ima konstantni člen $1 \Rightarrow f$ je obrnljiv

$\Rightarrow I = \mathbb{Z}[X]$

Torej I ali vsebuje $\langle X \rangle$ ali pa je od ko se x dodela

praideali: $\langle 0 \rangle$, maksimalni: $\langle X \rangle, \langle p \rangle$

$(X^2 - 6), \ldots$ neskončno

npr $(X-6)$ ni praideal

$\mathbb{Z}_{(2)} = \{\frac{a}{b} \in \mathbb{Q} \mid b \text{ lih}\}$

~~ideali~~ $\{\frac{a}{b}; a \text{ sod. } b \text{ lih}\} = \langle 2 \rangle$

ende: $\{\frac{a}{b}; a, b \text{ lih}\}$

torej je $\langle 2 \rangle$ edini maximalni

praideali: $\langle a \rangle, \langle 2 \rangle, \langle 2^a \rangle$

vsi ideali so vsebovani v

$\mathbb{Z} \times \mathbb{Z}$: vsi ideali $n\mathbb{Z} \times m\mathbb{Z}$ $n, m \in \mathbb{N}_0$

maksimalni $p\mathbb{Z} \times \mathbb{Z}$, $\mathbb{Z} \times p\mathbb{Z}$

(1)

a) (p) ... maksimalen

Recimo da $(p) \subseteq (a) \Rightarrow p = ax \Rightarrow x \in R^\times \vee a \in R^\times$

$x \in R^\times \Rightarrow (p) = (a)$ $\quad a \in R^\times \Rightarrow (a) = R$

$\bigcap_{i=1}^{\infty} (p_i) = (a)$

$\forall i: (p_i) \supseteq (a) \Rightarrow a = p_i m$

PID $\Rightarrow$ UFD $\Rightarrow$ $a = p_1 \cdot \ldots \cdot p_n$ končno mnogo

torej ~~končno~~, $a = 0$

~~$Nil(I) = \sqrt{I}$~~

$Nil(R) = \sqrt{I}$

$a^n = 0 \Rightarrow p_1^n \cdot \ldots \cdot p_k^n = 0$

ker n-deljivost ni ... je $Nil(R) = 0$

b) $\mathbb{Z}_{(p)} = \{ \frac{a}{b} ; p \nmid b \}$

maximalni ideali

$(\frac{a}{b}) = \{ \frac{\alpha}{\beta} \frac{a}{b} \}$

$p \nmid a \Rightarrow \beta = a \; \alpha \cdot \beta \rightsquigarrow 1 \in (\frac{a}{b})$

$p \mid a \Rightarrow$ ✓

$\left(\frac{px}{b}, \frac{py}{c}\right) = \left\{ \frac{\alpha}{\beta} \frac{px}{b} + \frac{\gamma}{\delta} \frac{py}{c} \right\} =$

$= \frac{\alpha p x \delta c + \dots}{\beta b \delta c} \}$ $p \mid$ vsak element

$\frac{p}{b}$ je maksimalen

$\left(\frac{p}{b}\right) \subseteq I \Rightarrow$

$\exists \frac{x}{y} \in I \; p \nmid x \Rightarrow \frac{y}{x} \frac{x}{y} \in I$

1

$\left(\frac{p}{2}\right) \cap \left(\frac{p}{6}\right)$

(p)

① $I, J \triangleleft R$ $\quad (J:I) = \{x : xI \subseteq J\}$

$(J:R) = J$

$(R:I) = R$

a) $\mathrm{Hom}(R/I, M) \cong \{m \in M;\ mI = 0\}$

b) $\mathrm{Hom}(R/I, R/J) \cong J:I/J$

c) $\mathrm{Hom}(\mathbb{Z}/(m), \mathbb{Z}/(n)) \cong \mathbb{Z}/\gcd(m,n)$

d) $\dim \mathrm{Hom}(\dots) = 2$

e) $\mathrm{Hom}(I, J) \cong (J :_R I) := \{g \in R;\ gI \subseteq J\}$

f) izračunaj če $(2\mathbb{Z} : 4\mathbb{Z})$, $(2\mathbb{Z} :_{\mathbb{Q}} 4\mathbb{Z})$ in opiši

$\mathrm{Hom}_{\mathbb{Z}}(4\mathbb{Z}, 2\mathbb{Z})$

g) ...

a) $\mathrm{Hom}(R/I, M) \cong \{m \in M : mI = 0\}$

$\varphi: f \longmapsto f(1+I)$

$(x \mapsto xm) \longleftarrow m : \psi$

Dobra definiranost

$m - n \in I$

$R \xrightarrow{f} M \quad \ker \varphi = \{m; xm = 0 \ \forall x \in R\} = I$

$\downarrow \quad \nearrow \bar{f}$

R/I

???

$\psi(\varphi(f)) = \psi(f(1)) = \psi(x \mapsto x \cdot f(1)) = x \mapsto f(x) = f$

Se kuga odrau

b) $\mathrm{Hom}(R/I, R/J) \cong \{m \in R/J \,;\, m \cdot I = 0\}$ (R/J-valued) $=$

$= \{x+J \,;\, x \in R \;\; (x+J)I = 0\}$

$\Leftrightarrow xI + JI = 0$

$\Leftrightarrow xI \subseteq J$

$\Leftrightarrow x \in (J:I)$

c) $\mathrm{Hom}(\mathbb{Z}/(n), \mathbb{Z}/(m)) \cong \mathbb{Z}/\gcd(m,n) \cong (n):(m)$

$(n):(m) = \{a \in \mathbb{Z} \,;\, a(n) \subseteq (m)\}$

$\{a \in \mathbb{Z} \,;\, an = gm \;\; g \in R\}$ $\quad d = \gcd$

$= \{a\mathbb{Z} \,;\, a\frac{n}{d} = z\frac{m}{d}\}$

$= \{a \in \mathbb{Z} \,:\, \frac{m}{d} \mid a\}$

$m' = \frac{m}{d}$

$\frac{(J:I)}{J} = \frac{(m')}{(m'd)} \cong \mathbb{Z}/d$

$\| $

$\{0, 1, \ldots, d-1\}$

d) $\dim \operatorname{Ham}\left(\frac{k[x,y]}{(x,y)}, \frac{k[x,y]}{(x^2,xy,y^2)}\right)$

$I = (x,y) \quad J = (x^2, xy, y^2)$

$(J:I) = \{p(x,y) \in k[x,y] : p \cdot (x,y) \subseteq (x^2,yx,y^2)$

$= (x^2,yx,y^2)$ in

$px \in (x^2,yx\,y^2)$

$py \in (x^2,yx,y^2)$

$p = \lambda_{00} + x\lambda_{10} + y\lambda_{01} \ldots \quad / \cdot x$

$p = \sum_{i,j} \lambda_{ij} x^i y^j \quad / \cdot x$

$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \lambda_{ij} x^{i+1} y^j$

$i,j > 0$ so elementi ideala in $j \geq 0$

$\lambda_{00} x$ ni element ideala

$\Rightarrow \lambda_{00} = 0$

$(J:I) = \{p$ dinami, ki imajo prosti člen $0\} = \{p : p(0,0) = 0\}$

$= (x,y)$

$\frac{(J:I)}{J} = \frac{(x,y)}{(x^2,xy,y^2)}$ osnovna sestavo $\lambda_1 x + \lambda_2 y$

$\dim(\ldots) = 2$

e) R cel

$\mathrm{Hom}(I, J) \cong (J :_k I) := \{ g \in K ; gI \subseteq J \}$

$\varphi : f \longmapsto \frac{f(x)}{x}$ zanek $x \in I - \{0\}$ $I \neq (0)$

$\psi : g \longmapsto (x \longmapsto gx)$

$\frac{f(x)}{(x)} \in (J :_k I)$

$\frac{f(x)}{x} y^{\in I} = \frac{\overbrace{f(x)y}^{\in J \cdot I}}{x} = \frac{f(xy)}{x} \overset{\in J}{\underset{\in I}{}} = \frac{f(y)x}{x} = f(y) \in J$

bijekcija?

$\varphi(\psi(g)) = \varphi(x \mapsto gx) = \frac{gx}{x} = g$

$\psi(\varphi(f)) = \psi(\frac{f(x)}{x}) = x \longmapsto \frac{f(x)}{x} y = f(y)$

b) $\mathrm{Hom}(R/I, R/J) \cong (J:I)/J$

$$(J:I)/J = \{r \in R;\ rI \subseteq J\}/J =$$
$$= \{r + J \in R/J;\ rI \subseteq J\}$$

Mje I modeel

$$\mathrm{Hom}(R/I, R/J) \cong \{m \in R/J;\ mI = 0^{\in M}\} \text{ (po a))} =$$
$$= \{x + J \in R/J;\ (x+J)I = 0 \in R/J\}$$
$$= \{x + J \in R/J;\ (xI + JI) \subseteq J\} =$$
$$= \{x + J \in R/J;\ xI \subseteq J\} = (J:I)/J$$

f) $(2\mathbb{Z} : 4\mathbb{Z}) \cong \underset{\mathbb{Z}}{Hom}(4\mathbb{Z}, 2\mathbb{Z}) \cong \mathbb{Z}$

$\cong\mathbb{Z} \quad \cong\mathbb{Z}$

na tablo

g)

$$(x,y)^{\vee} = \underset{\parallel}{Hom}(\underset{\parallel}{(x,y)}, k[x,y]) \cong k[x,y]$$

$\quad\ I \qquad\ J$

$(J:I) = (k[x,y] :_k (x,y)) =$

$= \{ \frac{f}{g} \in K(x,y) ; \frac{f}{g} \cdot (x,y) \subseteq k[x,y] \}$

$\frac{f}{g} \cdot x \in k[x,y]$ in

$\frac{f}{g} y \in k[x,y]$

$\rightsquigarrow fy = gh_2$

$x \mid f \wedge x \mid g$ ne, lerso že podrejsnani

$\frac{f}{g} x = h_1$

$fx = h_1 g$

↑ pradelement

$x \mid h_1 \vee x \mid g \rightsquigarrow g \in (x)$ $\Leftrightarrow x \mid h_2$

$\Downarrow$

$h = x h_1'$

$\Rightarrow f = h_1' g$

$\frac{f}{g} = h_1' \in k[x,y]$

???

2. Ugotovi, ali je kratko eksaktno zaporedje razcepno

a) $0 \to \mathbb{Z}_{(m)} \xrightarrow{n \cdot _} \mathbb{Z}_{mn} \xrightarrow{\text{modulo } n} \mathbb{Z}_{(n)} \to 0$

b) $0 \to \frac{\mathbb{C}[X]}{(X^k)} \xrightarrow{X^{l-k} \cdot _} \frac{\mathbb{C}[X]}{(X^l)} \xrightarrow{\text{kanon.}} \frac{\mathbb{C}[X]}{(X^{l-k})} \to 0$

c) $0 \to \mathbb{Z} \xrightarrow[k \mapsto (k+k+\dots)]{} \mathbb{Z} \oplus \prod_{\mathbb{N}} \mathbb{Z}_{(n)} \to \prod_{\mathbb{N}} \mathbb{Z}_{(n)} \to 0$

$(x, (a_i)) \mapsto (x, a_1, a_2, \dots)$

$\uparrow \mod n$

a) $\mathbb{Z}_{(m)} \oplus \mathbb{Z}_{(n)} \cong \mathbb{Z}_{(mn)}$

$\varphi: \mathbb{Z}_n \to \mathbb{Z}_{mn} \in \mathrm{Hom}\left(\mathbb{Z}_n, \mathbb{Z}_{mn}\right) \tilde{\cong} \frac{mn\mathbb{Z} : n\mathbb{Z}}{(mn)}$

$\varphi(x) = kx$ (sledi iz ciklicnosti

$k \in \frac{mn\mathbb{Z} : n\mathbb{Z}}{(mn)} \cong \frac{m\mathbb{Z}}{(mn)}$

$\varphi(x) = kx = k'mx \; : \; k' \in \mathbb{Z}$

vsi homomorfizmi so take oblike

$\pi \circ \varphi = id$

$\pi(\varphi(x)) = \pi(k'mx) = k'mx \mod n$

$k'mx = x \mod n \iff$

$k'm1 = 1 \mod n \iff k'm = 1 \mod n$

$\uparrow$ ali obstaja tak k

ja če $\gcd(m, n) = 1$

① R ring, M R-modul $I, J \trianglelefteq I$

a) $\mathrm{Ham}(R/I, M) \cong \{m \in M;\ Im = 0\} = S$

$\varphi: f \longmapsto f(1+I)$ $\quad f(1+I) \in S$

$a \in I \quad af(1+I) = f(a+I) = f(I) = 0$

$\psi: m \longmapsto (x+I \longmapsto xm)$

Dobra definiranost ψ?

$x - y \in I \quad xm - ym = (x-y)m = 0 \Rightarrow xm = ym$

$\varphi(\psi(m)) = \varphi(x+I \longmapsto xm) = m$

$\psi(\varphi(f)) = \psi(f(1+I)) = x+I \longmapsto xf(1+I) = x+I \longmapsto f(x+I)$

$= f$

b) $\mathrm{Ham}(R/I, R/J) \cong (J:I)/J$

$(J:I)/J = \{r \in R;\ rI \subseteq J\}/J =$

$= \{r + J \in R/J;\ rI \subseteq J\}$

Mje I modeel

$\mathrm{Ham}(R/I, R/J) \cong \{m \in R/J;\ mI = 0 \overset{\in M}{\}}$ (po a)) =

$= \{x + J \in R/J;\ (x+J)I = 0 \in R/J\}$

$= \{x + J \in R/J;\ (xI + JI) \subseteq J\} =$

$= \{x + J \in R/J;\ xI \subseteq J\} = (J:I)/J$

c) $Hom(\mathbb{Z}/n, \mathbb{Z}/m) \cong \mathbb{Z}/gcd(m,n)$ $\quad m,n \geq 1$

$I=(n) \quad J=(m)$

$Hom(\mathbb{Z}/I, \mathbb{Z}/J) \cong (J:I)/J$

$(J:I) = \{x \in \mathbb{Z} . \; x(n) \subseteq (m)\} =$

$= \{x \in \mathbb{Z} . \; \forall k \in \mathbb{Z} . \exists c \in \mathbb{Z} . \; xkn = mc\}$

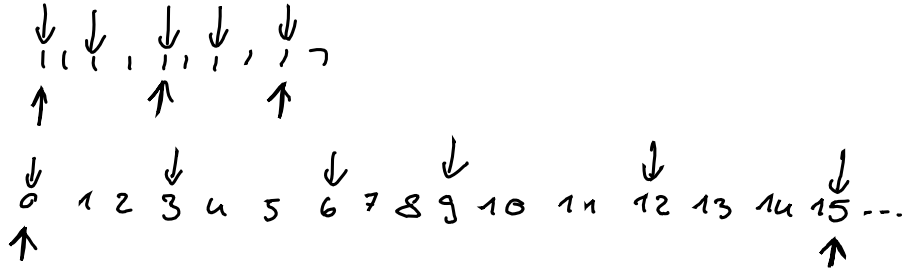
$\Leftrightarrow \{x \in \mathbb{Z} . \; \exists g \in \mathbb{Z} . \; xn = mg\}$ $\quad d = gcd(n,m)$, $n = n'd$, $m = m'd$

$(J:I) \cong \{x \in \mathbb{Z} . \exists g \in \mathbb{Z} . \; xn' = m'g\}$

$\cong \{x \in \mathbb{Z} . \; m' | xn'\}$

$\cong \{x \in \mathbb{Z} . \; m' | x\} = (m')$

$\frac{J:I}{J} = \frac{(m')}{(dm')} = \{0, 1 \ldots d\} = \mathbb{Z}/d$



d) $\dim_k \mathrm{Hom}_{k[x,y]}\left(k[x,y]/(x,y),\ k[x,y]/(x^2,xy,y^2)\right)$

k polje.

$$I=(x,y) \qquad J=(x^2,xy,y^2)$$

$$\mathrm{Hom}(\dots)\cong (J:I)/J$$

$$(J:I)=\left\{p\in k[x,y];\ \begin{matrix} p\cdot x\\ p\cdot y\end{matrix}\rightsquigarrow \in (x^2,xy,y^2)\right\}$$

$$p=a_{00}+a_{10}x+a_{01}y+a_{11}xy+\dots \quad /\cdot x$$

$$p\cdot x=a_{00}\cdot x+\underbrace{a_{10}x^2+a_{01}xy+a_{11}x^2y+\dots}_{\in (x^2,xy,y^2}$$

a_{00} mora biti 0

$$(J:I)=\{p: p(0,0)=0\}$$

$$(J:I)/J=\{p+(x^2,yx,y^2)\ ;\ p(0,0)=0\}$$

$$p=a_{10}x+a_{01}y+\underbrace{a_{11}xy+a_{02}x^2+\dots}_{\in (x^2,y^2,xy)}$$

$$=\{a_{10}x+a_{01}y+(x^2,yx,y^2)\ ;\ a_{10},a_{01}\in k\}$$

$$\Rightarrow \dim\left((J:I)/J\right)=2$$